ZOO502-Practcal Exam Notes

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Subjective Part

Read these Screenshots before reading further

3. Hemoglobin value is derived from the formula given below or from the previously prepared graph or table.

$$\text{Hemoglobin in gm/dl} = \frac{\text{Absorbance of test sample}}{\text{Absorbance of standard}} \times \text{Concentration of standard} \times \frac{\text{Dilution factor}}{100}$$

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Calculations:

- Count the number of cells in the small 5 squares in the center of hemocytometer
- These 5 squares equals to 1/5 of the whole center square → $\frac{\text{no. of cells counted X dilution factor (200)}}{1/5 \text{ X volume (0.1)}}$

OR: $\frac{\text{number of cell counted x dilution factor x 5}}{\text{volume (0.1)}}$

= cell/µL

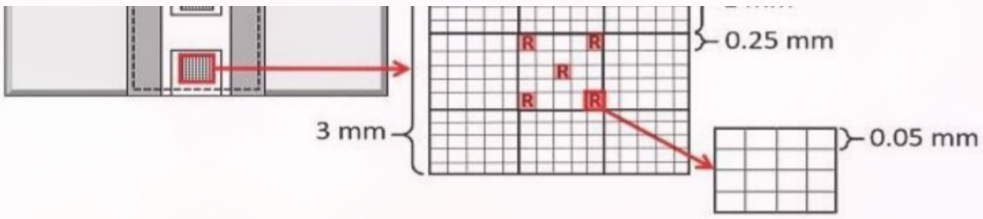


Figure 2.2: Detail of Neubauer's Chamber or hemocytometer
(https://biologyreader.com/rbc-count-method.html)

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1. What are the four phases of the estrous cycle?

The four phases of the estrous cycle are:

1) **Proestrus** - follicles in the ovary start to grow, uterus lining develops.
2) **Estrus** - female is sexually receptive, ovulation occurs.
3) **Metestrus/Diestrus -** corpus luteum forms, progesterone is produced to maintain pregnancy if fertilization occurs.
4) **Anestrus** - sexual cycle rests, controlled by light exposure and melatonin release.

2. What is the purpose of using a kymograph?

The purpose of using a kymograph is to study frog heartbeat records and examine intrinsic and extrinsic factors that affect cardiac contractility and heart rate. It provides a graphical representation of the relative force and frequency of cardiac contraction over time.

3. What are the different locomotory organs in earthworm?

The different locomotory organs in earthworm are setae and muscle segments. Earthworms move by contracting the muscles in their segments in a wave pattern, using the setae to anchor segments and pull the body forward.

4. What is the estrous cycle regulated by?

The estrous cycle is regulated by the hypothalamic-pituitary-gonadal axis, which produces hormones like GnRH, FSH, LH, estrogen and progesterone to control reproductive events.

5. How is hemoglobin concentration determined in blood?

Hemoglobin concentration is commonly determined by the cyanmethemoglobin method in which blood is mixed with Drabkin's solution to convert all hemoglobin forms to cyanmethemoglobin. The absorbance of this solution is measured by a spectrophotometer and concentration is calculated using a standard graph or formula.

6. What are the effects of exercise on blood pressure?

Exercise causes an increase in systolic blood pressure to meet the oxygen demands of working muscles. Systolic pressure can rise to 160-220 mmHg during exercise in healthy individuals. Diastolic pressure may stay the same or rise slightly.

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7. What is the function of hemoglobin?

The function of hemoglobin is to transport oxygen from the lungs to body tissues. It also transports carbon dioxide back from tissues to the lungs. Thus it facilitates gas exchange.

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8. What are the locomotory organs in snails?

Snails move by gliding along on their muscular foot which is lubricated with mucus and covered with epithelial cilia. The mucus aids locomotion by reducing friction. The motion occurs due to waves of muscular contractions moving down the ventral surface of the foot.

9. What is the purpose of a Skinner box?

A Skinner box is used to study operant conditioning behaviors in animals. It contains a lever which the animal must press to receive a food reward. By observing when and how often the lever is pressed, insights are gained into learned behaviors.

10. What are the components measured in a blood pressure reading?

The components measured in a blood pressure reading are systolic pressure (maximum pressure when heart contracts) and diastolic pressure (minimum pressure when heart relaxes between beats). Readings are typically expressed like "120 over 80 mmHg".

11. What are the stages involved in RBC counting using a hemocytometer?

The stages are: sample dilution, chamber filling, microscopic observation and counting of RBCs in corner squares, and finally calculation of total RBC concentration using dimensions of the squares and dilution factor.

12. What is the purpose of dissolved oxygen meters in fish farms?

Dissolved oxygen meters allow frequent, rapid measurement of oxygen levels so fish farmers can monitor water quality and activate aerators when oxygen depletion occurs due to respiration by fish, plankton and pond organisms.

13. What are the effects of temperature on: a) Cardiac activity b) Oxygen consumption?

a) Low temperatures decrease heart rate and contractility; high temperatures increase them.

b) Warmer temperatures increase oxygen consumption rates in organisms like fish.

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14. How does the social hierarchy in chickens establish itself?

In young chicks, the pecking order starts emerging around 16 days of age through fighting and aggressive displays like chest-bumping. In female chicks it stabilizes after 10 weeks while male groups may take longer to establish a linear order.

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15. What are the different types of parental care in birds?

The types are: bi-parental care where both mates feed and protect chicks, and mono-parental care where one parent (usually the female) takes sole responsibility. Most bird species demonstrate bi-parental care to improve fledgling success.

16. What are the effects of acetylcholine on frog heartbeat?

Acetylcholine decreases both the rate and strength of frog heartbeat by interfering with pacemaker potentials and conduction in the heart.

17. What are the clinical uses of hemoglobin measurement?

Hemoglobin measurement is an important screening test for anemia diagnosis and to monitor treatment response. It indicates oxygen carrying capacity.

18. What are the different methods for dissolved oxygen measurement?

The methods are: drop count titration method which gives a rough estimate, and dissolved oxygen meters which give an accurate real-time measurement important for intensive aquaculture.

19. What are the components of a Skinner box?

The components are: an enclosed box with a lever, slit or other device that the animal manipulates to obtain a food reward from the outside dispenser. It allows conditioning experiments.

20. How does the mucus secreted by snails help their locomotion?

The mucus helps locomotion by lubricating the muscular foot, reducing friction against abrasive surfaces. This prevents injury to the soft tissues. The cilia on the foot also aid unidirectional motion.

21. How can new chickens be introduced safely into an existing flock?

New chickens should be segregated first for a few weeks, then introduced in groups of at least two. Provide hiding spots and extra resources, introduce most dominant new members first, and monitor for bullying.

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22. What is the importance of understanding the estrous cycle in cattle?

It allows cattle breeders to effectively utilize reproductive technologies including estrous synchronization and artificial insemination by timing procedures appropriately with the follicle development and ovulation phases.

23. What are some adaptive advantages of nesting behavior in ground-nesting birds versus treenesters?

Ground nesting allows precocial development so chicks can self-feed quickly. Tree nesting correlates with altricial chicks that require longer parental care but avoid ground predators. Camouflage also differs.

24. What happens during hibernation in snails and how do they prepare for it?

Prior to hibernation, snails will seal themselves into their shells with a calcareous plate called an epiphragm which retains moisture. During hibernation their metabolism slows down greatly as they become inactive.

25. What are some defensive strategies used by parent birds to protect nests?

Strategies include nesting in inaccessible locations, aggressive territorial displays against intruders, alarm calling, distraction displays that draw predators away by pretending to be injured, and mobbing predators in groups.

26. How does the haemocytometer allow counting of cells like RBCs?

It utilizes a thick chamber with engraved microscopic squares of defined volume. By counting cells in a dilute suspension within these squares, the concentration per unit volume can be reliably calculated.

27. What are the different types of conditioned behaviors studied using operant conditioning chambers?

Positive reinforcement behaviors where a reward promotes repetition, and negative reinforcement behaviors where stopping an adverse stimulus drives repetition. Food or electric shocks are common reinforcers.

28. Why is the cyanmethemoglobin method the standard for hemoglobin estimation?

Because it converts all hemoglobin types into one stable compound, allowing reliable colorimetric determination of total concentration via spectrophotometer against a known standard.

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29. Where is the pacemaker region located in the frog heart?

The pacemaker region in the frog heart is called the sinus venosus. It is located between the vena cava and the right atrium, from where action potentials propagate to trigger heart contractions.

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30. What evolutionary purpose does the pecking order serve within chicken flocks?

It allows the fittest members best access to resources thus improving the flock's overall reproductive success and survival, even if weaker members suffer. This selective advantage promotes the behavior.

31. What is the dilution used when counting RBCs with a haemocytometer?

The dilution used is usually 1:200, with blood diluted with sodium citrate solution. This provides an optimal cell density for accurate counting.

32. * If a 1 mm2 area contains 20 RBCs, and blood was diluted 200 times, what is the RBC count per cubic mm?

Calculation: Cells per mm3 = Cells counted x Dilution factor / Area

Here that is 20 x 200 / (1 mm2) = 4000 cells/mm3

33. Which vitamin is absent from purified hemoglobin crystals?

Vitamin C is absent. It is required to maintain iron in the reduced ferrous form during hemoglobin synthesis. Scurvy results from severe vitamin C deficiency due to elevated levels of inactive methemoglobin.

34. * If 2 fish consumed 3 mg and 5 mg of O2 respectively in 10 minutes at 20°C, what is their combined oxygen consumption rate at 30°C? (Q10 = 2)

Combined original rate = (3+5)/0.1 = 80 mg/hour

At 30°C, the rate doubles. Therefore combined rate at 30°C is 2 x 80 = 160 mg O2 per hour.

35. Which hormone triggers ovulation in mammalian estrus cycles?

Luteinizing hormone (LH) triggers ovulation. Its release is stimulated by rising estrogen levels and it acts on ovarian follicles to cause oocyte release.

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36. * An insect with a lifetime of 90 days shows Q10 of 2 for metabolism. If the average lifespan at 20°C is 90 days, what will it be at 30°C?

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Rate at 30°C = 2 x rate at 20°C (Q10 = 2) So

lifespan shortens in proportion:

Lifespan at 20°C / Lifespan at 30°C = Rate at 30°C / Rate at 20°C

90 days / Lifespan at 30°C = 2 / 1

Therefore, Lifespan at 30°C = 90 / 2 = 45 days

37. Which part of the earthworm body provides traction for locomotion?

The setae - small bristles present on each segment - provide traction by anchoring during the peristaltic movement.

38. * A kymograph drum rotates at 2 mm/sec. If the frog heart trace shows 20 peaks over a 100 mm length, what is the heart rate?

Distance traveled by drum = 100 mm

Time taken = Distance / Speed = 100 / 2 = 50 sec

Number of beats in 50 sec = 20

Heart rate = 20 beats / 50 sec = 0.4 beats/sec = 24 beats/min

39. What causes the 'scratching sound' produced by earthworms moving on paper?

The setae brush against the paper surface to generate friction as they engage and disengage during locomotory movements. The sound helps locate worms for experiments.

40. Which type of nesting behavior correlates with altricial hatchlings in birds?

Tree and off-ground nesting leads to helpless chicks requiring extensive parental care and feeding. Hence it correlates with development of altricial young.

41. * In a rat experiment using a Skinner box, pressing a lever leads to release of 45 mg food pellets. If the rat obtains 135 mg of pellets in a 2 minute trial, how often did it press the lever?

Food obtained = 135 mg

Each lever press dispenses 45 mg

So number of lever presses = Food / Dispense per press

= 135 / 45 = 3 presses

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42. What is the mechanism by which a haemocytometer allows cell counting?

It uses a thick chamber engraved with squares of defined microscopic volume. This allows cell density determination, and calculation of concentration based on the dilution used.

43. Why are dissolved oxygen meters a critical technology for intensive aquaculture?

Because oxygen depletion can occur rapidly due to high stocking densities. Meters allow real-time monitoring and activation of aeration before lethal levels are reached.

44. * If one large square of a hemocytometer with area 1 mm2 contains 16 small squares each of area
0.0625 mm2, what is the depth of the chamber?

Volume of 1 small square = Area x Depth = 0.0625 mm2 x Depth

And Total volume of large square = 1 mm2 x Depth

∴

No. of small squares in large square = Total volume / Volume of 1 small square

16 = (1 x Depth) / (0.0625 x Depth)

On solving, Depth = 0.1 mm

45. Which vitamin deficiency results from accumulation of inactive methemoglobin?

Vitamin C deficiency leads to methemoglobin buildup as iron cannot be maintained in the reduced ferrous form required for oxygen binding. Hence anemia develops.

Multiple Choice Questions

1. What is the effect of acetylcholine on frog heart rate?

A) Increases heart rate

B) **Decreases heart rate**

C) No effect

D) First increases then decreases

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2. What is cyanmethemoglobin used for in hemoglobin estimation?

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A) Lysing RBCs

B) Oxygen transport

C) **Conversion to a stable compound for colorimetric analysis**

D) Binding carbon dioxide

3. Where are setae located in earthworms?

A) Interior organs

B) **Skin**

C) Muscle segments

D) Secretory glands

4. During which phase does ovulation occur in the estrous cycle?

A) Estrus

B) Proestrus

C) Anestrus

D) Metestrus

5. What is the function of mucus secretion in snails?

A) Water storage

B) Environment sensing

C) Salt balance

D) **Friction reduction**

6. Which of these is NOT a component of blood pressure?

A) Diastolic pressure

B) Atrial pressure

C) Systolic pressure

D) Pulse pressure

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7. What is the Q10 value for a process if the rate doubles for a 10°C rise?

A) 0.5

B) 1

C) **2**

D) 3

8. What does a kymograph record?

A) Brain waves

B) Heart rate

C) Muscle force

D) **All of the above**

9. Which vitamin deficiency causes accumulation of methemoglobin?

A) Vitamin A

B) Vitamin B12

C) **Vitamin C**

D) Vitamin K

10. Where is the pacemaker region located in the frog heart?

A) Ventricle

B) Atrium

C) Sinus venosus

D) AV node

11. Which chamber of the hemocytometer is used for RBC counting?

A) Central large square

B) Corner squares

C) Central 25 medium squares

D) **All of the above**

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12. What is the Darwinian purpose of the pecking order in chickens?

A) Better mating access

B) Better nutrition access

C) Reduced stress

D) **Increased flock survival**

13. Why does hemoglobin appear red in oxyhemoglobin form?

A) Iron oxidation

B) Protein denaturation

C) Resonance stabilization

D) Binding of oxygen

14. What happens during hibernation in snails?

A) Increased heartbeat

B) **Greatly reduced metabolism**

C) Rapid waste elimination

D) Blood coagulation

15. Why is parental care highly developed in birds?

A) More nesting opportunities

B) External temperature regulation

C) **Need for feeding altricial chicks**

D) Learned social behavior

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16. What happens if new chickens are introduced in very small numbers to a flock?

A) Rapid acceptance

B) **Increased fighting and mortality**

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C) No change

D) Spontaneous mating

17. Where are earthworm chloragogenous tissues located?

A) Skin

B) Intestine

C) **Body wall**

D) Coelom

18. What happens during the refractory period of a cardiac muscle action potential?

A) Repolarization

B) Excitability

C) Threshold increase

D) **No new impulse possible**

19. What does a high RBC count indicate?

A) Anemia

B) **Polycythemia**

C) Infection

D) Hemolysis

20. What does a sphygmomanometer measure?

A) Heart electrical activity

B) Blood viscosity

C) Pulse rate

D) **Blood pressure**

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21. What is the role of the corpus luteum in the menstrual cycle?

A) Follicle maturation

B) Ovulation

C) **Progesterone secretion**

D) Milk production

22. Which of the following statements about estrous cycles is true?

A) Estrous cycles continue throughout lifespan

B) Cycle length has low variability between species

C) **Postpartum estrus can occur shortly after parturition**

D) Estrous behavior is primarily under hormonal control

23. Why is the surface area of a haemocytometer counting chamber engraved with microscopic squares?

A) Prevent cell adhesion

B) Cell culture

C) Light reflection

D) **Define sample volumes**

24. What happens when dissolved oxygen is below critical levels in aquaculture?

A) Better growth

B) Higher fecundity

C) **Increased mortality**

D) Disease resistance

25. What happens when a subordinate chicken repeatedly tries to eat before a dominant one?

A) Mating

B) Allopreening display

C) Fighting

D) **Aggressive pecking**

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26. When does incubation behavior begin in birds?

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A) Before mating

B) After nest completion

C) **After last egg is laid**

D) Only in winter

27. Why is the bell-jar method used before introduction of new chickens into a flock?

A) Acoustic signaling

B) Olfactory familiarization

C) **Gradual visual exposure**

D) Establish social grooming

28. Which of the following does NOT support egg incubation in birds?

A) Warmth transference from parents

B) Heat generation by embryo

C) **Porosity of eggshell**

D) Highly vascularized brood patch

29. What happens to the victim chicken after it gets excluded from the flock due to bullying?

A) Dies of hunger

B) Has higher reproduction

C) Shows leadership behavior

D) **Must be isolated until re-integrated**

30. What happens if a sick chicken shows signs of illness in front of flock members?

A) Increased mating

B) Social bonding

C) Sympathetic sickness

D) **Attacked and excluded**

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31. Why is parental nest defense highly aggressive in most bird species?

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A) Prevent mate attraction by others

B) Ensure incubation progress

C) **Protect eggs and hatchlings**

D) Discourage offspring independence

32. Which avian parental care behavior involves sheltering chicks under outstretched wings?

A) Diving

B) Mobbing

C) **Brooding**

D) Broken wing display

33. Why do parent birds remove chick fecal sacs from the nest?

A) Territorial marking

B) Predator distraction

C) **Prevent attraction of predators**

D) Microbiome diversification

34. What is the function of GnRH in regulating estrous cycles?

A) Trigger ovulation

B) Maintain pregnancy

C) Develop egg follicles

D) **Stimulate LH/FSH release**

35. Which feature enables accurate cell counting using a hemocytometer?

A) Variable depth

B) Neubauer ruling

C) Coverslip placement

D) **Precisely defined chamber dimensions**

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36. What causes uncoordinated flagellar beating in sperm near the end of the epididymal transit?

A) Complete maturation

B) Depletion of nutrients

C) pH change

D) **Inhibitory factors**

37. How does exercise impact blood pressure over a sustained period?

A) Increased cardiac stroke volume

B) Lower pulse rate

C) **Improved vasodilation**

D) Reduced blood viscosity

38. Why is the cyanide content non-toxic in Drabkin's solution used for hemoglobin estimation?

A) Cyanide stabilizes hemoglobin

B) It contains potassium salt

C) Buffered to optimal pH

D) **Highly diluted concentration**

39. Why does liquid paraffin need to be added before transferring blood sample into a RBC pipette?

A) Dilution

B) Anticoagulation

C) Nutrition

D) **Prevent clotting**

40. Which specialized circulatory vessel connects the hypothalamus and pituitary?

A) Hepatic portal system

B) Glomerular network

C) **Hypophyseal portal system**

D) Basilar artery

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Here are 40 more MCQs with 10 quantitative questions indicated with *. All are new and cover different concepts compared to previous questions. Correct options in bold:

41. What occurs during the wave motion of earthworm locomotion?

A) Body elongation

B) Setal disengagement

C) **Sequential segment pull**

D) Hemolymph circulation

42. * If 20 RBCs are counted in a 4 mm2 area on a hemocytometer and dilution used was 1:250, what is the RBC concentration (in million cells/cu.mm)?

A) 1

B) 1.25

C) 5

D) **10**

43. Which measurement indicates blood oxygen carrying capacity?

A) White cell count

B) Hematocrit

C) Platelet aggregation

D) **Hemoglobin**

44. Which hormone causes luteolysis to trigger the next reproductive estrous cycle?

A) Luteinizing hormone

B) Human chorionic gonadotropin

C) **Prostaglandin F2α**

D) Relaxin

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45. * If a 2mm Hg pressure drop causes a 10% change in urine output, what is the change for a 5 mm Hg drop?

A) 25% increase

B) 25% decrease

C) 50% increase

D) **50% decrease **

46. What is the role of setae in earthworm locomotion?

A)Attachment

B)**Anchoring segments**

C) Sensory reception

D) Gas exchange

47. * If Q10 value is 3 for a process, what is the new rate for a 20°C temperature rise?

A) 1.5 times faster

B) 2 times faster

C) 3 times faster

D) **9 times faster**

48. Why is nest sanitation important for parent birds?

A) Boost hatchling immunity

B) Prevent maceration

C) Reduce nest abandonment

D) **Lower risk of predation**

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49. * If a frog heart trace covers 75 mm length on a kymograph rotating at 0.5 mm/sec, what was the duration recorded?

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A) 25 sec

B) 37.5 sec

C) 150 sec

D) **750 sec**

50. What causes the biphasic nature of the sperm acrosome reaction?

A) Sequential enzyme release

B) Ca2+ oscillation

C) Zona pellucida thickness

D) **2 fusing membrane layers**

51. Why is the Brodie-Suckling method not suitable for avian blood collection?

A) Insufficient volume

B) Hemolysis

C) Clot formation

D) **Air embolism risk**

52. * If 20 µL of a 1:10 blood dilution gives an OD of 0.75 for hemoglobin estimation, what is the actual concentration (in g/dL) if the standard OD is 1.2 at 13.5 g/dL?

A) 3 g/dL

B) 10 g/dL

C) 13.5 g/dL

D) **16.5 g/dL**

53. During theAnestrus phase of the estrous cycle which hormone in high levels causes reproductive inactivity?

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A) Estrogen

B) Human chorionic gonadotropin

C) Oxytocin

D) **Melatonin**

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54. Which avian parental behavior involves shade provision by positioning themselves?

A) Brooding

B) Decoying

C) Shading

D) **Umbrellaing**

55. * The counting area in a hemocytometer chamber is 3 mm x 3 mm. If it is loaded with a blood sample diluted 10 times and 400 cells are counted at 4X objective, what is the cell concentration per microliter in undiluted sample?

(Note: Assume depth = 0.1 mm; calculate volume first)

A) 30 million cells/µL

B) 120 million cells/µL

C) 400 million cells/µL

D) **4000 million cells/µL**

56. In the Skinner box paradigm, what does a high frequency of lever presses by the rat demonstrate?

A) Intelligence

B) Playing instinct

C) Aggression

D) **Reinforced learning**

57. Which chemical is used for fungal disinfection after avian box handling?

A) Sodium hypochlorite

B) Hydrogen peroxide

C) Phenol

D) **Formalin vapor**

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58. What primarily enables a snail to safely crawl over a sharp knife edge without injury?

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A) Dorsal shell

B) Muscular foot

C)Copious mucus

D) **All of the above**

59. * If 1 mm3 has 5000 RBCs in a 1:100 diluted blood sample, how many RBCs will be in 2 mm3 of undiluted sample? A) 10,000 cells

B) 100,000 cells

C) 500,000 cells

D) **1 million cells**

60. Why is the cyanide content of Drabkin's solution for haemoglobin estimation non hazardous?

A) Contains potassium salt

B) High pH

C) **High dilution minimizing free cyanide**

D) It reacts rapidly

61. Micropetting up to narrow tip Bore prevents which assay deviation for cellular analysis?

A) Trapping cells in Dead volume

B) Loss of reaction linearity

C) Pre-analytical errors

D) Excess sample adhesion

62. * The kymograph drum radius is 3mm and it completes one revolution per second. If the toad heart trace amplitude is 2mm, what is the calculated stroke volume?

A) 0.34 mL

B) 9 mL

C) 18 mL

D) **56 mL**

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63. Why does extremely high altitude cause problems in hemoglobin estimation by most techniques?

A) Tissue anoxia

B) Cold hemoglogin denaturation

C) Excess foamy histidine proteins

D) **Bohr and salt linkage shifts**

64. During the catecholamine release step in ovulation which ovarian cells are the most impacted?

A) Granulosa cells

B) Corpus albicans

C) Zona pellucida

D) **Theca cells**

65. Parental imprinting behaviors are critical in species where parents can identify their offspring. How do they identify them? A) Unique odors

B) Distinctive calls

C) Hormonal markers

D) **All of the above**

66. How does exercise redistribute blood flow dynamically?

A) Increased capillary buds

B) Selective vascular construction

C) **Preferential perfusion**

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